Reliability-aware active learning for FFLO phase spaces

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Finite-momentum superconducting phase diagrams require repeated minimization over pairing amplitude and momentum, followed by spectral tests wherever topology is meaningful. We combine an exact Bogoliubov–de Gennes oracle with active learning that schedules, but never replaces, these calculations. The method reconstructs the two-dimensional Normal, uniform-superconducting, and FFLO regions while evaluating 6,880 of 77,361 prescribed candidate locations, an exact-location sampling fraction of 8.89% (11.24-fold reduction). Restricting topology to reliable gapped FFLO states reveals connected conventional and topological FFLO support separated by a numerically reconstructed gap-closing region. Extending the same procedure to $(k_BT/t, J_A/t, \mu/t)$ exposes systematic thermodynamic and topology-aware structure across a finite sampled domain. The latter evidence is reported as discrete support and fixed-μ slices, not as a converged continuous topological surface or complete three-dimensional atlas.

INTRODUCTION

Superconductivity with nonzero center-of-mass pairing momentum provides a route beyond a spatially uniform condensate. Fulde and Ferrell and, independently, Larkin and Ovchinnikov established the finite-momentum pairing and spatially modulated mechanisms now grouped under the FFLO setting [1, 2]. Subsequent reviews have developed the thermodynamic, material, and ultracold-gas consequences of Fermi-surface mismatch and competing pairing momenta [3–5]. The central phase-selection problem is therefore intrinsically thermodynamic: competing minima must be compared over both pairing amplitude and momentum.

The search for FFLO structure spans several physical platforms. Heavy-fermion and organic superconductors have displayed field-induced phases and spectroscopic, calorimetric, or magnetic signatures discussed in relation to FFLO order [6–10]; these representative studies motivate the problem without implying that every identification is universal or uncontested. Population-imbalanced ultracold Fermi gases provide a complementary tunable setting for pairing under mismatch, phase separation, and finite-momentum phases [11–15].

Altermagnets add a distinct route to this competition. Their compensated magnetic order can nevertheless generate symmetry-structured momentum-dependent spin splitting [16–19]. Model studies have connected this splitting to zero-field finite-momentum or other unconventional pairing states [20–22], and neighboring altermagnetic constructions have been proposed for topological superconductivity and Majorana modes [23, 24]. These precedents motivate an altermagnetic finite-q problem but do not determine the phase structure of the one-dimensional model studied here.

Topology adds a second, logically separate requirement. Finite-momentum pairing can coexist with topological superfluidity in spin-orbit-coupled models [25–27]. More generally, bulk symmetries and invariants organize topological superconducting phases [28–32]. Majorana-platform studies connect those bulk distinctions to boundary physics [33–36]. In a gapped one-dimensional Bogoliubov–de Gennes (BdG) system in symmetry class D, the relevant $\mathbb{Z}_2$ distinction can be expressed through a Pfaffian invariant [37, 38]; its use is meaningful only after bulk-gap applicability is established. A topology-resolved FFLO map must therefore identify a reliable thermodynamic minimum before classifying trusted gapped FFLO states.

This hierarchy is computationally demanding because every accepted parameter point requires minimization of $F(\Delta, q)$, comparison with the normal reference, adaptive checks of shallow or edge minima, and, where applicable, full-zone spectral and Pfaffian diagnostics. Active learning and Bayesian experimental design formalize the selection of informative sequential evaluations [39–41]; phase-diagram applications demonstrate boundary-focused sampling [42–44], while adaptive sampling has also reduced expensive electronic-structure, atomistic, and chemical search budgets [45–50]. For a physical phase diagram, however, an acquisition model should decide where to calculate next, not replace the exact calculation that defines the phase.

Here we combine an exact BdG oracle with active learning to resolve the thermodynamic and topology-aware structure of an altermagnetic finite-momentum superconductor. The oracle supplies every physical label, while the surrogate redirects subsequent evaluations toward uncertain or rapidly varying regions. This strategy reconstructs the two-dimensional thermodynamic phases using 8.89% of the prescribed candidate locations, resolves connected cFFLO and tFFLO support within the reliable gapped FFLO sector, and exposes systematic structure across a finite three-dimensional domain. The three-dimensional evidence remains a discrete point cloud and fixed-μ slices rather than a converged continuous topological surface.

RESULTS

Exact oracle and active-learning scheduler.— For external coordinates $x = (k_B T/t, J_A/t, \mu/t, \ldots)$, the exact calculation minimizes the normal-referenced free energy $F(\Delta, q)$ constructed from the BdG spectrum. The lowest numerically reliable minimum defines a Normal, uniform-superconducting, or FFLO state. Minima close to a scanned q edge trigger expansion of the momentum window, whereas near-zero-Δ competition triggers a refined comparison with the normal state. Phase identity is stored separately from numerical status. Only thermodynamically accepted, reliable, and gapped FFLO states are passed to the topological calculation. The surrogate then selects the next microbatch of coordinates; it neither supplies a phase label nor replaces the exact calculation (Fig. 1). Complete equations, tolerances, algorithms, network architecture, and acquisition rules are given in the Supplemental Material.

Two-dimensional thermodynamic reconstruction.— The two-dimensional search starts from a sparse random sample of the complete candidate grid. As the exact dataset grows, subsequent evaluations concentrate around the Normal–superconducting and uniform-SC–FFLO structures while retaining exploratory coverage (Fig. 2). The primary discovery run evaluates 6,880 unique coordinates out of 77,361 prescribed candidate locations, corresponding to an 8.89% sampling fraction and an 11.24-fold reduction in exact locations relative to exhaustive evaluation of that grid. Later iterations change the reconstructed phase regions and their principal boundaries only weakly. Numerically delicate evaluations are retained separately rather than promoted to an additional phase; phase counts, return accounting, diagnostic histories, and operational stopping settings are reported in the Supplemental Material.

Topology-resolved FFLO structure.— Topology is evaluated only after thermodynamic acceptance and only for reliable gapped FFLO observations. A full-Brillouin-zone minimum-gap calculation establishes applicability, and the signs of the Pfaffians at $k = 0$ and π assign cFFLO ($\nu = 0$) or tFFLO ($\nu = 1$). The discrete exact observations support connected topology-resolved structure (Fig. 3). A contour reconstructed within the sampled support and local opposite-$\mathbb{Z}_2$ brackets guide the eye toward the gap-closing region; neither is an exact phase boundary or a third stable phase. Changing the interpolation neighborhood from $k = 8$ to $k = 6$ or 12 produces a small but measurable variation in the derived contour. We therefore use it only as a guide, and do not claim that all possible nodal alternatives or refinements have been exhausted.

Higher-dimensional exploration.— The same oracle–scheduler separation extends to $x = (k_B T/t, J_A/t, \mu/t)$ over $\mu/t \in [-0.5, 1.5]$. The exact three-dimensional point cloud reveals systematic thermodynamic and topology-aware structure, and four narrow fixed-μ bins expose how that structure evolves across the third coordinate (Fig. 4). The slices contain exact observations only, with no interpolation between them. This support is sufficient to demonstrate higher-dimensional exploration, but not to establish a converged continuous topological surface or a complete three-dimensional atlas.

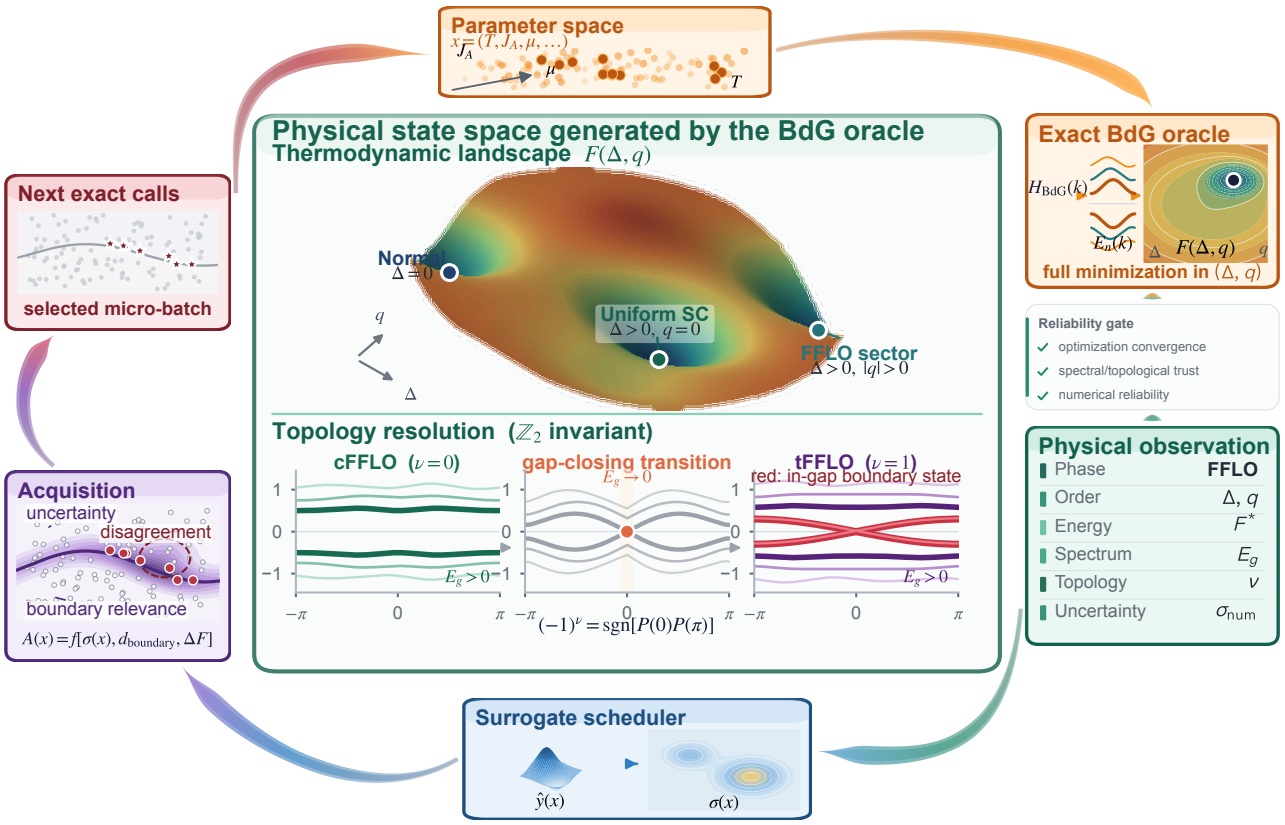


FIG. 1. **Exact physics and adaptive scheduling.** Representative minima of $F(\Delta, q)$ distinguish Normal, uniform-SC, and finite-q FFLO states. A full-zone gap and Pfaffian invariant distinguish cFFLO, a gap-closing transition region, and reopened tFFLO only within the reliable gapped FFLO subset. The surrounding surrogate and acquisition cycle schedules subsequent exact calls and does not determine physical labels.

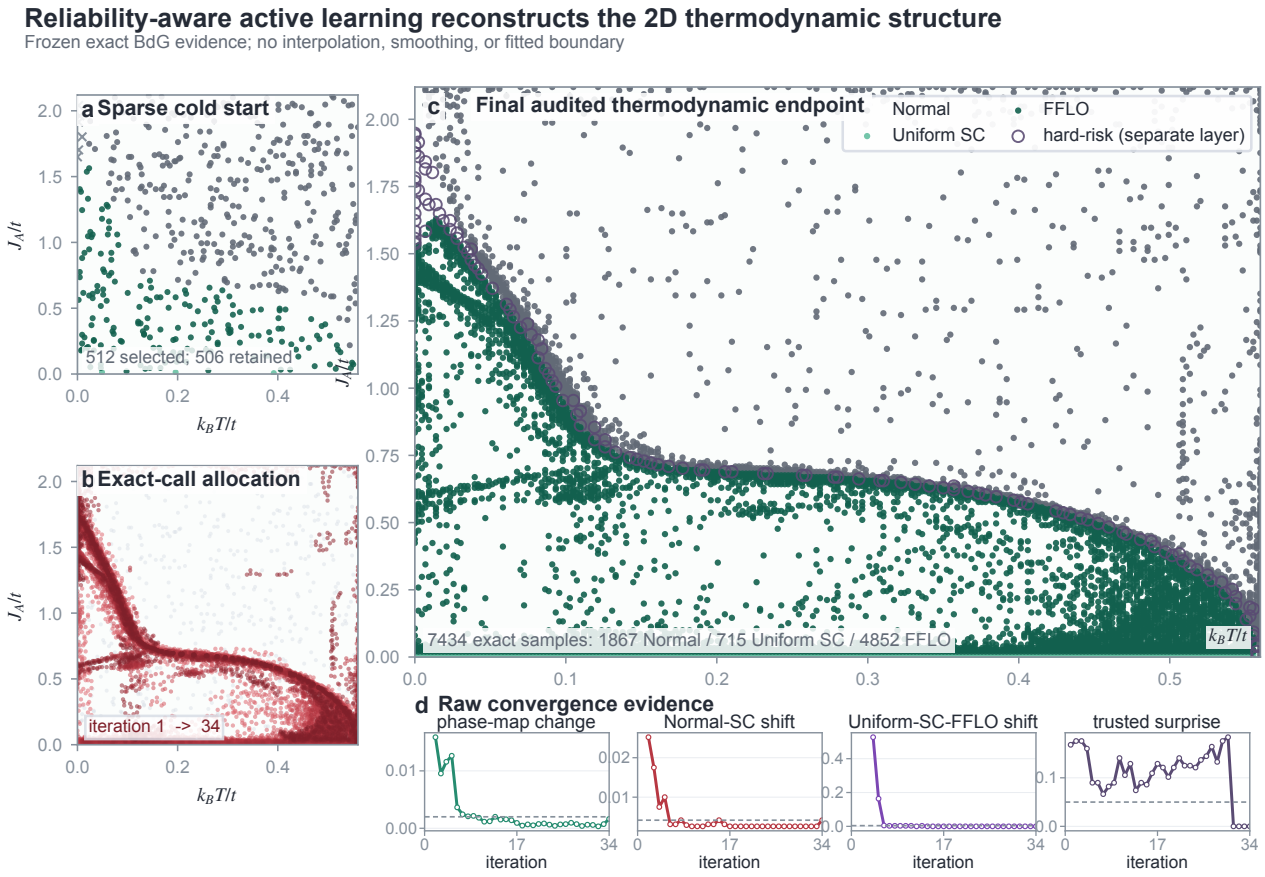


FIG. 2. **Adaptive reconstruction of the two-dimensional thermodynamic phases.** **(a)** Sparse random-grid initialization. **(b)** Iteration-resolved exact-call allocation. **(c)** Final exact-sample map of the Normal, uniform-SC, and FFLO regions; open symbols mark numerically flagged evaluations rather than a phase. **(d)** Numerical histories used to monitor changes in the reconstructed map and boundaries. Complete counts and operational settings are provided in the Supplemental Material.

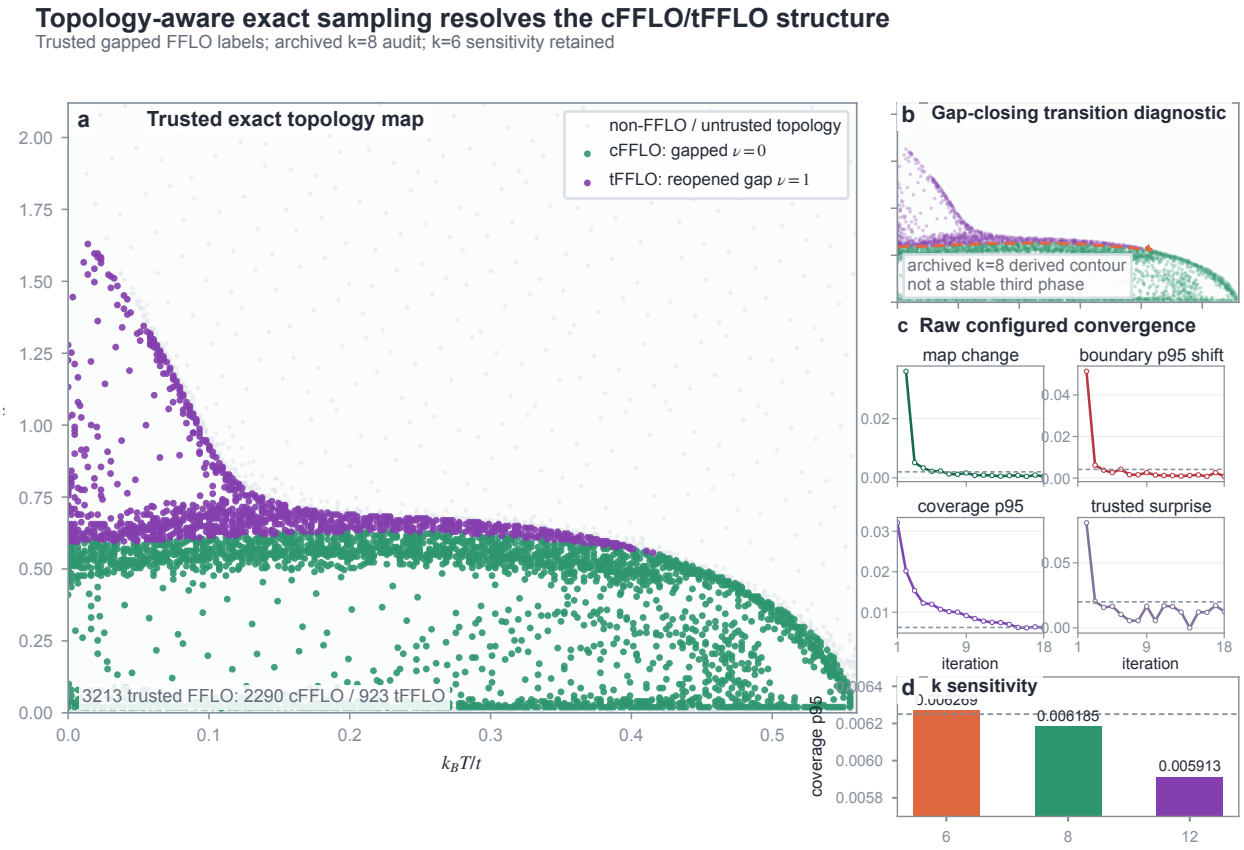


FIG. 3. **Topology-resolved cFFLO and tFFLO structure.** **(a)** Reliable gapped FFLO observations classified by the full-zone gap and Pfaffian invariant. **(b)** A support-restricted reconstruction and local opposite-$\mathbb{Z}_2$ brackets indicate the gap-closing region but not an exact boundary. **(c)** Raw numerical histories. **(d)** Neighborhood sensitivity for $k = 6, 8$, and 12; the finite variation motivates treating the contour as a diagnostic guide.

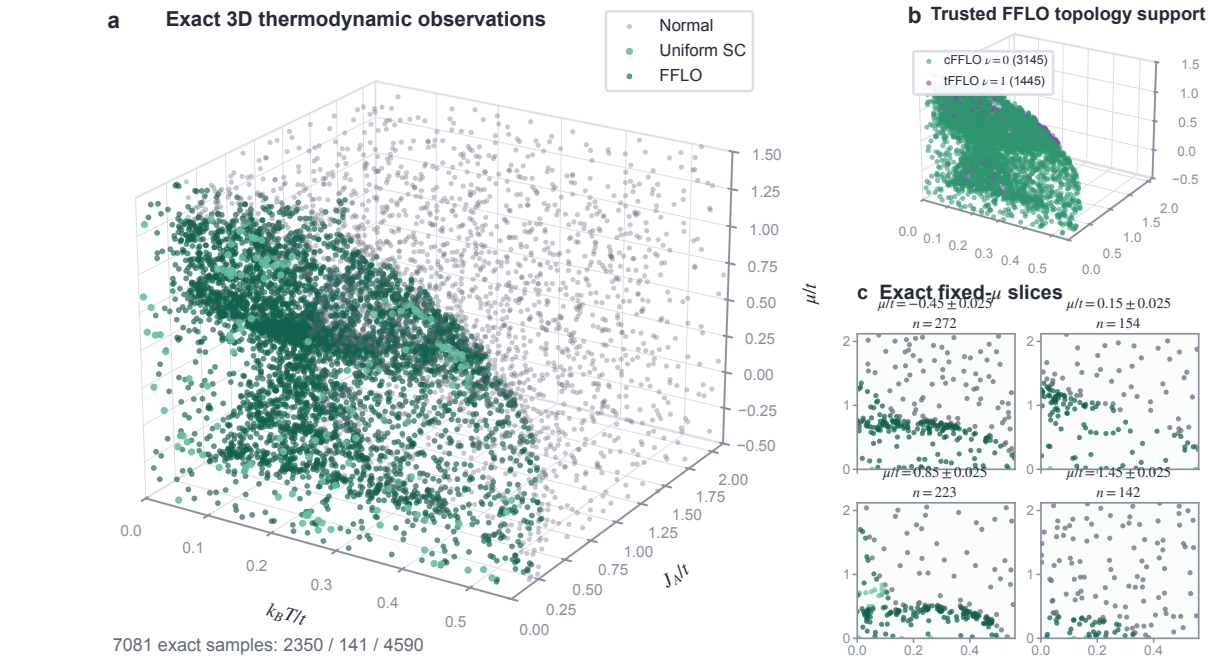


FIG. 4. **Higher-dimensional active-learning exploration.** **(a)** Exact thermodynamic observations in $(k_BT/t, J_A/t, \mu/t)$ over a finite chemical-potential interval. **(b)** Discrete reliable FFLO topology support. **(c)** Four fixed-μ bins shown without interpolation between slices. The data reveal three-dimensional structure but do not define a continuous topological surface or complete atlas.

DISCUSSION

Reliability-aware scheduling changes where exact calculations are spent without changing what constitutes a physical result. This separation is essential near shallow thermodynamic minima, momentum-window edges, and gap closings, where a surrogate prediction alone cannot establish a phase. The two-dimensional calculations show that a small fraction of the prescribed candidate grid can recover the principal thermodynamic structure, and the topology-resolved calculation shows how spectral applicability can be layered onto that result without conflating a gap-closing diagnostic with a stable phase. The three-dimensional extension demonstrates the value of selective exact sampling as the parameter domain grows, while its discrete support sets a clear limit on the claim.

Exploratory learned-acquisition studies trade local topology support against sequential surprise, but neither tested policy has an overall advantage and no candidate node is validated as a new phase. These studies motivate future adaptive discovery policies rather than a claim of learned-acquisition superiority.

A longer-term direction is a learned physical world model that represents thermodynamic state, spectra, and calibrated uncertainty jointly. The present evidence does not yet validate the spectral descriptors or expected-information-gain policy required for such a model, so that direction remains an outlook rather than an established method.

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